

Energy Management System

1. Overview

China's huge population and rapid pace of economic growth make its energy needs particularly challenging. After two decades of annual growth averaging eight percent, it is now the world's second largest energy consumer after the United States, according to the International Energy Agency (IEA). China's National Bureau of Statistics reports consumption of coal in China rose 18.6 percent to 3.06 billion tons in 2009, outpacing supply, which rose 18 percent to 2.74 billion tons from its 2006 equivalent. (See Table 1-1 *Energy consumption and supply (2006-2009)* below). As a result, the government is facing a series of challenges to expand its energy supply while increasing efficiency.

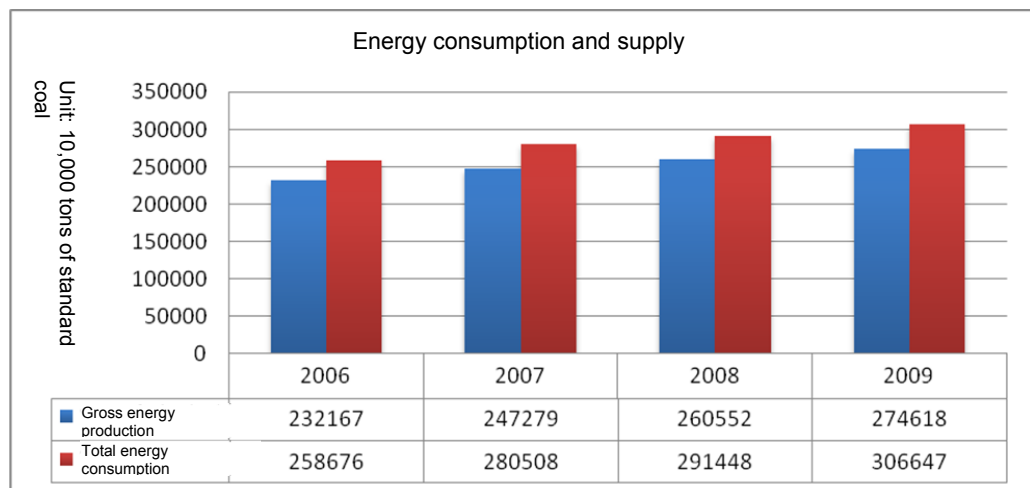


Table 1-1: Energy production vs. energy consumption (2006-2009)

The bulk of the demand comes from the industrial sector, which is often hampered by shortages as rising consumption puts pressure on domestic energy reserves. In 2011, China's major industrial provinces, Guangdong, Guangxi, Guizhou, Yunnan and Hainan, suffered from the worst power shortages since 2006. In May 2011, power regulators were forced to impose "off-peak " periods in the Pearl River Delta industrial area of Guangdong province, affecting industrial towns located in the area, namely Dongguan, Shenzhen and Foshan, in an attempt to reduce power consumption.

To meet year-end targets, 60 percent of small factories in Dongguan for instance, only received electricity three days a week, while companies in Baoan and Shenzhen had six days of electricity supply, as opposed to seven full days. Similarly, factories in 21 towns in Dongguan, including Shijie, Tanxia and Dalang only operate four days a week. These severe shortages combined with the rising value of the Yuan have raised costs for local businesses and depleted their profit margins.

China is at a highly energy-intensive stage of growth, consuming nearly 10 percent of the world's energy resources. But government intervention through the introduction of energy saving schemes presents opportunities to enhance efficiency in the industrial sector. Companies, the majority of which are SMEs,

will benefit from implementing such schemes to help them reduce power usage, increase energy efficiency and reduce costs.

The following chapters explain the benefits of implementing energy management systems and offers best practice advice to help SMEs use energy more efficiently.

ENERGY MANAGEMENT SYSTEM STANDARDS: ISO50001 AND GB/T23331-2009

China's Standard Certification Center (CSC) began research into improving energy management systems and upgrading national standards in 2002. In 2009, the government implemented its newly developed research GB/T 23331-2009, "Requirements for Energy Management Systems." Since then, China has made regular updates to its systems to match national standards with those of the International Organization for Standardization (ISO), which published its latest international energy management standards, the ISO 50001, in June 2011.

Organizations across the world face energy-related challenges, including those related to energy supply, reliability and climate change matters. The ISO 50001 is a framework that helps companies manage their energy systems and plan better to save energy and to reduce pollution as well as costs. ISO estimates these standards can reduce global energy consumption by 60 percent. The ISO 50001 provides the following benefits:

- » Resolves energy efficiency problems
- » Improves energy usage of energy-consuming assets
- » Estimates environmental impact of greenhouse gases;
- » Improves energy management and communication;
- » Provides best practices for energy efficiency;
- » Prioritizes new energy-saving technology;
- » Improves energy efficiency of supply chains; and
- » Details greenhouse gas reduction plans

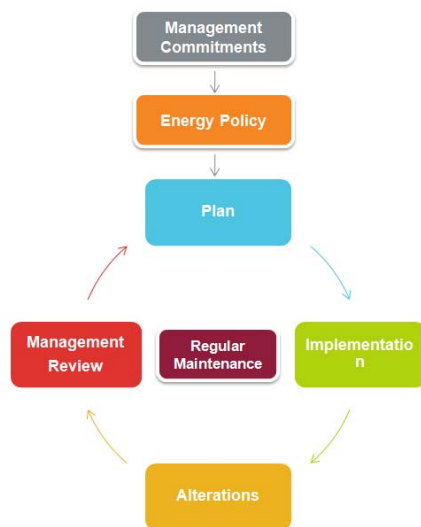


Figure 1-2: Energy Management Model

Both ISO50001 and GB/T23331-2009 are based on the same model (See Fig.1-2 Energy management model), which applies the “Plan, Do, Check and Action” cycle (PDCA). This handbook provides China-based companies, particularly SMEs, with an overview of ISO 50001 as it relates to the Chinese GB/T23331-200 standard.

2. Establishing an Energy Management System

2.1 MANAGEMENT COMMITMENT

Both international and Chinese energy management standards require the support of senior managers to commit to facilitating energy efficiency throughout their organizations. These include commitments to establish, implement, maintain and improve energy management systems on a regular basis.

Specifically, managers must commit in writing to:

- » Integrate energy policies and regulations into company operations;
- » Incorporate energy targets into overall business strategies;
- » Provide staff with energy management training;

- » Conduct management reviews of energy management processes; and
- » Keep energy-related equipment well maintained.

Energy Saving Commitments

- » Establish a system to collect, analyze, and report data related energy consumption, and ensure correctness and integrity of that data;
- » Designate personnel with expertise and technical experience, or above, to be in charge of energy management, and file this personnel in government agencies in charge of energy saving;
- » Enhance measuring instrument management and equip and use qualified measuring instruments according to relevant regulations;
- » Establish a responsibility system and award organizations and individuals with good performance in energy-saving activities;
- » Make and implement energy-saving plans and technical measures;
- » Comply with national energy limitation standards and make corporate standards that are stricter than national standards;
- » Use energy efficiency markings on products according to relevant regulations;
- » Conduct energy-saving education and training regularly;
- » Never produce, import, or sell any products or equipment (or use equipment) that are clearly specified as "to be eliminated" by government agencies or those which do not conform with energy efficiency standards;
- » Never provide free energy to employees or implement any "package fee" system for energy use.

These commitments should be printed and posted in public areas in the factory to promote energy management.

A good energy manager should display the following competences:

- » A senior/middle class manager who can report to the top management directly;
- » Be familiar with main production processes;
- » Have a certain understanding of production equipment, electrical, boiler, air-conditioning, lighting, and auxiliary systems;
- » Have a certain degree of financial knowledge and energy management; and
- » Strong management ability and is able to implement improvement plans independently.

2.2 ENERGY POLICY (OPTIONAL)

National and international energy management standards encourage companies to create energy policies specific to their company operations. But implementation of these policies is up to the discretion of managers and companies.

Good energy policies:

- » Are integrated with the company's business activities, products and services;
- » Reduce energy consumption, increase efficiency and guarantee regular system maintenance;
- » Comply with applicable laws and regulations;
- » Provide a framework for making and evaluating energy targets and indicators;
- » Offer employees training materials so that they can understand and implement energy policies; and
- » Are readily available to all relevant parties within the organization.

2.3 ENERGY MANAGER

Senior executives must assign an independent specialist to oversee energy management within the company, with the exception of SMEs, whose directors

EXAMPLE- Energy management organization chart

Leader:
Vice General Manager

Members:
Production Supervisor
Equipment Engineer
Financial Supervisor

are allowed to designate an internal energy manager from their middle-level management staff. In this case, the standards recommend appointing a supervisor or higher-level executive from the production, maintenance or engineering divisions.

The energy manager is required to:

- » Establish, implement, maintain, and improve energy management systems in accordance to the national and international standards;
- » Provide regular updates to top management on all energy management systems;
- » Submit proposals for improvements; and
- » Liaise with external organizations on energy management issues.

2.4 PLANNING

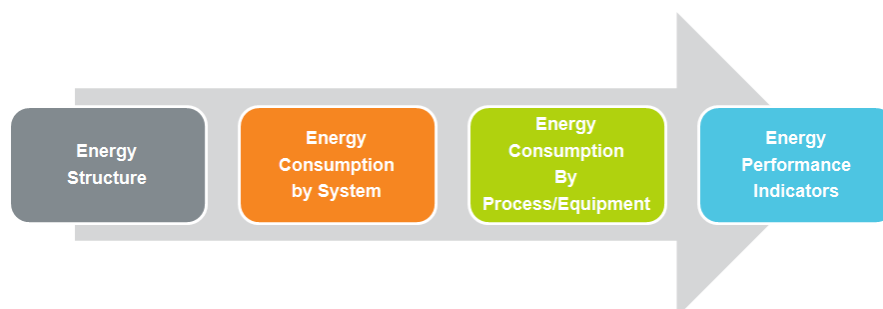
The energy manager is responsible for developing an action plan in line with national and international standards, the first step in the “Plan, Do, Check and Action” cycle.

Any plan must:

- » Identify key energy performance indicators;
- » Adhere to rules, regulations and other requirements that affect management systems;
- » Highlight energy benchmarks;
- » Set energy targets; and
- » Design an efficient energy management platform

2.4.1 IDENTIFICATION OF ENERGY FACTORS

Identifying key energy performance indicators is vital for the planning process, as



it provides managers with a clear picture of how their company uses energy and can highlight ways to manage resources better.

The manager's first step is to determine the energy consumption “structure” of his or her company, in other words, what energy resources does the company need to run its operations, be it gas, coal or electricity (see example below).

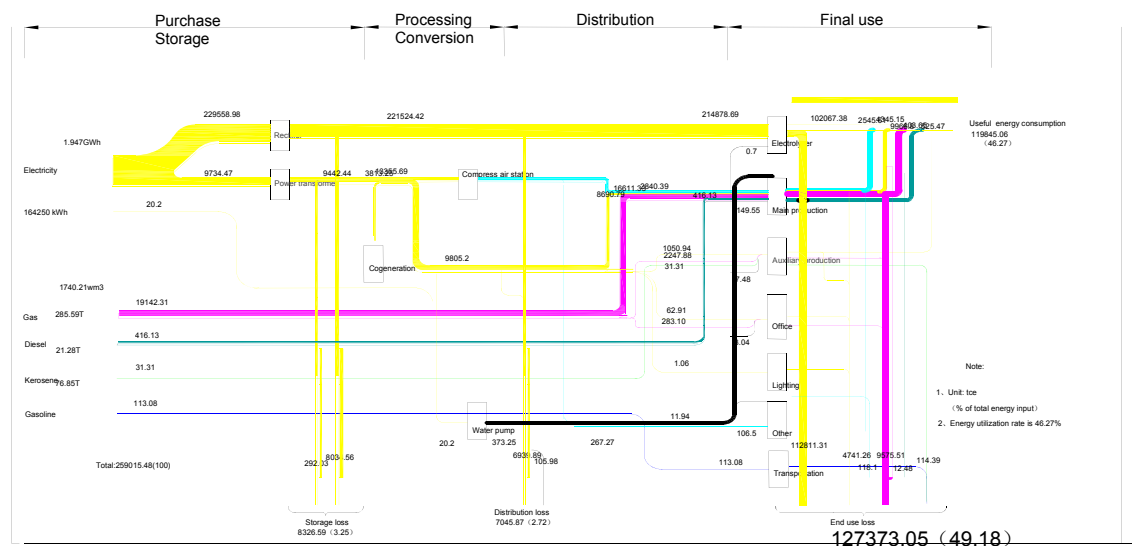
Energy Consumption of Company A

Item	Natural gas (10,000 m3)	Kerosene (t)	Diesel oil (t)	Power purchased (10,000 kWh)	Power generated (10,000 kWh)	Compressed air (10,000 m3)	Surface water (10,000 t)	Total energy consumption (tce)	Recycled (tce)	Actual consumption (tce)
Actual amount	1711.52	14.80	297.54	182825.66	2780.84	62.34	164.06	—	—	—

Conversion to standard coal	10832.21	21.78	433.55	224692.74	3417.65	22.44	200.81	239621.18	3640.91	235980.27
%	4.52	0.01	0.18	93.77	1.43	0.01	0.08	100.00	1.52	98.48

Next a manager should identify the basic systems within their companies that consume energy, such as air conditioners, lighting systems, refrigerators, etc. Then, managers need to look at manufacturing processes that consume energy. Once overall energy use is defined, the manager should design an energy flow chart (see below) based on their overall assessment of the company's energy consumption. The chart should include a detailed analysis of energy usage through the company, highlighting which equipment consumes the most energy. To correctly make this analysis, energy managers need feedback from employees, including factory line managers and technical supervisors.

Fig 1-3 Energy flow



The goal of this process is to allow an energy manager to identify the company's key energy performance indicators, provide top management with a prognosis of the company's energy consumption, and offer recommendations on energy-saving technology.

After identifying the key performance indicators, managers can apply the following principles:

- » Ensure compliance with the relevant laws and regulations that apply;
- » Ensure a safe working environment;
- » Technical feasibility;
- » Economic feasibility;
- » Energy saving;
- » Prioritize low-cost/no-cost solutions.

Table: Energy Performance Indicator

Energy consumption process	Energy consumption asset/equipment	To be eliminated by law or not	Status	Energy factor	Technical feasibility 1-10	Economical feasibility 1-10	Adverse impact (safety, quality, environment) High, Medium, Small, No	Assessment result A: maintain B: control with high priority C: to be studied D: no measure	Related division
Air conditioning	Central air conditioner	No	No control by area, time, and climate	Seasonality	9	9	No	B	Office building Workshop
				Heat/cool air demand varies in different area and time	7	8	No	B	

2.4.2 LAWS, REGULATIONS, AND OTHER REQUIREMENTS

Managers should update their energy performance indicator table on a regular



basis to make sure it's in line with the latest energy regulations and to help in updating targets for their company's future energy strategy. They should also keep a library of applicable rules and regulations.

List of laws and regulations on energy management systems

No.	Laws, regulations, and requirements	Issued by	Date of issuance and implementation
1	Energy Conservation Law	Standing Committee of the National People's Congress	Issued on October 28, 2007 and implemented on April 4, 2008
2	Measures for the Supervision and Administration of Energy Measurement	State Administration of Quality Supervision	Issued on July 22, 2010 and implemented on November 11, 2010
3	Measures for the Electricity Demand-Side Management	National Development and Reform Committee, Ministry of Industry and Information Technology, National Energy Administration, Ministry of Finance, State-owned Assets Supervision and Administration Commission, and State Electricity Regulatory Commission	Issued on April 11, 2010 and implemented on January 1, 2011
4	General Principles for Equipping and Management of Energy Measuring Instruments in Energy-consuming Units	State Administration of Quality Supervision	Issued on June 2, 2006 and implemented on January 1, 2007
5	Requirements on Energy Management System	State Administration of Quality Supervision	Issued on March 11, 2009 and implemented on November 1, 2009

6	Notice of the State Council on Further Strengthening the Elimination of Backward Production Capacities	State Council	Issued and implemented on February 6, 2010
7	The First List of Outdated Mechanical and Electrical Equipment (Products) with High Energy Consumption	Ministry of Industry and Information Technology	Issued on December 4, 2009
8	Renewable Energy Industry Development Guidance Catalog	National Development and Reform Committee	Implemented on November 29, 2005
9	Technical Guiding Catalog of Cleaner Production of National Key industry	State Environmental Protection Administration	Implemented on November 27, 2006
10	List of Recommended Energy-saving Mechanical and Electrical Equipment (Products) (1st batch)	Ministry of Industry and Information Technology	Issued on May 27, 2009
11	List of Recommended Energy-saving Mechanical and Electrical Equipment (Products) (2nd batch)	Ministry of Industry and Information Technology	Issued and implemented on August 16, 2010
12	Measures for Supervision and Management of High Energy-consuming Special Equipment	State Administration of Quality Supervision	Issued on July 3, 2009 and implemented on September 1, 2009
No.	Standard	Issued by	Date of issuance and implementation
1	Technical guides for evaluating the rationality of heat usage in industrial enterprise (GBT3486-1993)	State Bureau of Technical Supervision	Issued on June 19, 1993 and implemented on June 1, 1994
2	General principle of energy audit on industrial and commercial enterprise (GBT17166-1997)	State Bureau of Technical Supervision	Approved on December 22, 1997 and implemented on October 1, 1998
3	Guides for the power transformers energy saving operation in factories and mines (GBT13462-2008)	State Bureau of Technical Supervision	Approved on May 27, 2008 and implemented on January 1, 2008

Benchmark for coal consumption per gram

A coal-fired power plant consumes 390g coal for each kWh electricity generated.

Benchmark: The Outline of China's Policies on Energy-saving Technologies set coal consumption to 360g per kWh electricity by 2010, compared to 392g in 2000, with power consumption rates coming down to 5.1% by 2020 from 6.28% in 2000.

2.4.3 SETTING ENERGY MANAGEMENT BENCHMARKS

Managers should set energy efficiency benchmarks that measure consumption and the number of energy consumption assets in operation. Such benchmarks allow companies to compare their energy management systems against best practices in China and abroad. The benchmarks also allow companies to compare their energy efficiency results against national standards, such as the clean production standard or China's energy-saving technologies policy. (Please see the example below of a benchmark on net coal consumption identified in China's Policy Outline for Energy Conserving Technologies).

2.4.4 ENERGY TARGETS AND INDICATORS

Energy managers that are aware of their company's energy performance rating are in a better position to track energy and water consumption, and set targets and indicators (Please see the example below). As a result, they can improve policies, identify areas that need improvement, and measure the investments required for facility upgrades.

2.4.5 ENERGY MANAGEMENT PLAN

Energy management plans should be prepared on the basis of expert opinion and suggestions from front-line employees, bearing in mind the:

Target: Reduce unit coal consumption of boilers

Indicator: Reduce coal consumption per ton by 5% in 2012 (with that of 2010 as baseline).

Measures: Install steam accumulator; check CO and CO₂ smoke levels and install combustion control device.

Division in charge: Engineering
Person in charge: Maintenance Supervisor

Investment: RMB 200,000
To be started by: 2012.5
To be completed by: 2012.7

- » Purpose of the plan
- » Measures to take and persons in charge
- » Timetable for the plan
- » Performance evaluation/assessment methods and indicators
- » Rules for adjusting the plan in case of emergency

2.5 IMPLEMENTATION AND EXECUTION

The second step of PDCA cycle is "Do", which means implementation of the plan, which includes:

- » Competence building and information sharing among employees;
- » Stakeholder participation (optional according to international guidelines);
- » Document and record control;
- » Operation control; and
- » Emergency preparedness and response

2.5.1 COMPETENCE BUILDING AND INFORMATION SHARING

Companies should ensure energy managers and any other staff members involved in the energy management process have the required management and technical skills to execute the plan. Providing all employees with energy management training will make them more aware of better energy consumption practices, and will allow companies to receive valuable feedback from front-line employees as well as staff proposals. Companies can also provide their employees with incentive policies and awards for helping to reduce energy consumption and improve energy saving practices.

2.5.2 STAKEHOLDER PARTICIPATION (OPTIONAL)

To ensure their company's energy performance is constantly improving, energy managers should involve a variety of stakeholders, including their employees, suppliers, customers, local government staff, and members of the local community, in establishing their energy management system. Furthermore, stakeholder participation can help the company establish mutual trust with local communities, comply with local regulations and reduce energy-related risks and conflicts. Larger companies often have a stakeholder participation scheme already in place, but smaller companies, given their size and financial limitations, typically do not.

To establish stakeholder involvement, SMEs should advertise their plans to establish an energy management system; explaining how the plan can benefit the local community, the government and their customers. This would help rally support from stakeholders and increase their involvement in the company's energy management policies, particularly by employees. Since, it could act as an incentive for staff to contribute to the company's energy-saving schemes.

2.5.3 DOCUMENT AND RECORD CONTROLS

Energy management system documents should be prepared according to the following guidelines:

- » Documents must be prepared by an energy manager;
- » Documents must be approved by authorized personnel before publishing;
- » Documents must show the date of issuance or date of revision and include

According to the Energy Conservation Law, units with large energy consumption must submit an annual report to government agencies in charge of energy saving activities.

Such reports should cover energy consumption, energy efficiency, whether targets are achieved, energy saving benefit analysis and energy saving measures.

information on the personnel who approved them;

- » Documents must be updated regularly, with invalid or abolished articles deleted and properly marked;
- » The latest documents must be available to relevant divisions and personnel; and
- » Documents must comply with national laws and regulations, covering all energy management system requirements.

Case example: A coking plant's coal requirements

Purchased coal must have good coking properties and conform to the following:

The ash content should be less than 10% and cannot exceed 12.45%.

The sulfur content should be less than 1.5%, or 2.5% in the case of rare coal such as fat coal.

The moisture should be less than 12%.

The phosphorus content of pig iron should be lower than 0.01% ~0.015%.

Case example: Energy rules of company X.

In the summer, office temperatures should be set at 26 °C, with an air volume of five m³/hp.

The temperature of workshops should be kept at 28 °C, with an air volume of 20 m³/h.p

There are two kinds of energy management records, and these include:

- » **Internal records**, which comprise records on equipment maintenance, energy statistics, energy factor registration forms, internal reviews, training, improvement plans, etc.
- » **External records**, which comprise the energy audit report, archives on key energy-consuming equipment, etc.

2.5.4 OPERATION CONTROLS

Companies should follow the operation controls of their energy management systems closely, particularly the following:

- » Incorporating sustainability policies into their product development with the goal of reducing energy consumption, improving energy efficiency, and using reusable resources without compromising product quality, safety, or cost.
- » Consider energy saving before purchasing equipment to avoid acquiring equipment that consumes excessively high amounts of energy. Also, by optimizing the operating condition of existing equipment, and conducting maintenance on a regular basis, companies can reduce damage to equipment that could contribute to higher energy consumption.
- » Create a company standard for purchasing energy, including a selection of preferred energy suppliers, measurements and verification of energy sources, transportation and storage, etc.
- » Companies should measure, record and evaluate the energy consumption of their production and service processes so that they can eliminate outdated processes.

2.5.5 EMERGENCY PREPAREDNESS

Companies should consider making emergency plans based on real life situations that could affect their energy systems.



2.6 INSPECTION AND CORRECTION

“Check”, the third step of the PDCA cycle refers to inspection and correction of energy management plans.

2.6.1 MONITORING AND ASSESSMENT

Monitor energy management results, as well as key performance indicators (KPI) based on the company's energy-efficiency benchmarks, using an energy assessment form such as the one below.

EXAMPLE: Energy-saving Statistical Form of Company A													
2010 Fodder Plant Energy Saving and Emission Reduction Statistical Form													
Company name:		Date:											
		Energy consumption per ton and production volume				Compared with that of 2009 (+/-)				Increased by (%)			
		Power consumption (kWh)	Coal consumption (ton)	Oil consumption (L)	Production volume (10,000 RMB)	Power consumption (kWh)	Coal consumption (ton)	Oil consumption (L)	Production volume (10,000 RMB)	Power consumption (kWh)	Coal consumption (ton)	Oil consumption (L)	Production volume (10,000 RMB)
Mixed fodder	Powder												
	Granular												
Condensed													
Additive premixed													
Average													
Person in charge:		Filled by:		Tel:									

2.6.2 MAKING AN IMPROVEMENT PLAN

Prepare an improvement plan based on monitoring and assessment results. The aim is to:

- » Correct non-conformities;
- » Take preventive measures against non-conformities; and
- » Assign a person in charge of following up and ensuring that improvements are implemented.

EXAMPLE: Non-conformity report of Company A (Corrective Actions Form)

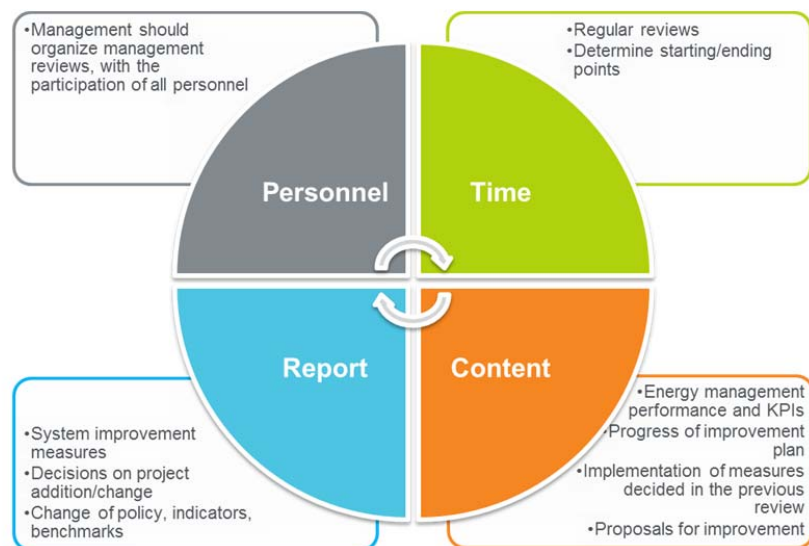
Division audited:	
Results	
The items above do not conform to _____ and are _____ non-conformities.	
Auditor/date:	
Receipt by manager of division audited:	Date:
Cause analysis and corrective actions:	
To be completed on:	
Manager of division audited:	Auditor: Approved by:
Implementation of corrective actions:	
Manager of division audited:	Date:
Effectiveness of corrective actions:	
Auditor/date:	

2.7 MANAGEMENT REVIEW

The last step of the PDCA cycle is "Action", which refers to the management review.

It should be conducted at an annual management review of the company's energy management system. The review should:

- » Include all personnel involved in energy management.
- » Look at how the energy management system has performed and examine KPIs; assess the progress of any improvement plans; ensure previous review measures were implemented, and include proposals for improvement.
- » Include a report detailing any improvements as well as proposals for changes or additions to the system, or changes in policies, performance indicators, or benchmarks.



Energy Audit

1. Overview

Energy audits compare the energy consumption and energy efficiency of different companies in the same industry by looking at comprehensive energy consumption, increase of energy consumption, and energy consumption per unit product etc.

Production at every company includes several processes and each process consumes energy. Thus, a company's production process is intrinsically linked to its energy consumption.

When examining a company's process energy consumption, it is necessary to analyze relevant factors, including the quality of raw materials, equipment used, type of energy used and subjective factors such as staff quality, production arrangements, and the condition of equipment.

In the case of processes that consume a large part of the overall energy used in production, it is necessary to conduct quantitative analysis on factors to find ways to resolve the issue.

1.1 WHAT IS AN ENERGY AUDIT?

Conducting regular energy audits, where auditors inspect, analyze and evaluate a company's energy consumption, allows energy managers to assess how much energy their company uses and to pinpoint opportunities for potential energy and cost savings. An audit is only useful, however, if energy managers can implement their auditors' recommendations.

1.2 CONTENTS OF AN ENERGY AUDIT

- » **Energy management systems** - review its compliance with national laws and regulations, the quality of its systems and policies and compare production levels to energy savings.
- » **Consumption and flow of energy** - accounts for energy input, storage, conversion, waste, sales and consumption.
- » **Statistics** - measures instruments, calibration, measuring rate, and instrument management. This also includes the scope of energy report, frequency of submission, breakdown level, depth of analysis, etc.
- » **Consumption indicators** - only accounts for a company's energy consumption, but not its energy inputs.
- » **Operating efficiency of equipment** – assesses to what extent company equipment makes efficient use of energy.
- » **Energy consumption during manufacturing** - includes consumption per unit of output value, per unit of product, direct and indirect energy consumption and energy consumption per comparable unit product.
- » **Energy cost indicators** - multiplies energy consumption by energy price.
- » **Energy savings** - differentiates actual energy consumption and benchmark consumption.
- » **Economic analysis of energy-saving projects** – makes a comparative analysis of energy costs for similar products and conditions.

1.3 ENERGY AUDIT LAWS AND REGULATIONS

- » Energy Conservation Law
- » Program for Energy Saving of One Thousand Enterprises
- » General Principles of Energy Audit on Industrial and Commercial Enterprise (GB/T17166—1997)
- » General Principles of Energy Monitoring Technology (GB/T15316)
- » General Principles for Calculation of Thermal Efficiency of Equipment (GB/T2588—1981)
- » General Principles for Calculation of Comprehensive Energy Consumption (GB/T2589—1990)
- » Guideline on Measuring and Testing Energy Consumption (GB/T6422—1986)
- » Energy Saving Calculating Methods (GB/T13234—1991)
- » Guideline on Energy Management in Industrial Enterprise (GB/T15587—1995)

Power Consumption Knowledge

A transformer consumes power even when there is no current running through it. The only way to avoid this is to turn off the power switch.

As long as there is a current, transformers, cables, and other electronic equipment consume power.

Stand-by Power Consumption

Most electrical equipment cannot be switched off completely without being unplugged, thus even if power is switched off, the equipment still uses electricity. A computer on standby mode consumes about 7.5W of power. If it is turned off, but the plug is not removed from the socket, it still consumes 4.81W of power.

The standby power consumption of a computer is worth 50kWh of consumption a year. Considering the amount of equipment in a factory, the potential to waste power is significant.

- » General Principles for Equipping and Management of Energy Measuring Instruments in Energy-consuming Units (GB/T17167—2006)
- » Technical guides for evaluating the rationality of heat usage in industrial enterprise (GB/T3486—1993)
- » Technical guides for evaluating the rationality of power usage in industrial enterprise (GB/T3485—1993)
- » Technical guides for evaluating the rationality of water usage in industrial enterprise (GB/T7119—1993)
- » Statistical method of energy balance in enterprises (GB/T16614—1996)
- » Methods of drawing up energy balance table in enterprises (GB/T16615—1996)
- » Methods of drawing energy network diagram in enterprises (GB/T16616—1996)

2. Key Reasons for an Energy Audit

2.1 CORPORATE ENERGY SAVING

Electricity is the most commonly used form of energy. If the use of this is managed correctly it can therefore help many companies cut their energy consumption and costs.

Lighting systems, electric motors and drive systems, and heating/cooling systems are the three most commonly used energy consuming systems across all industries, whether large or small. Thus, similar energy-savings strategies can be applied to any size business.

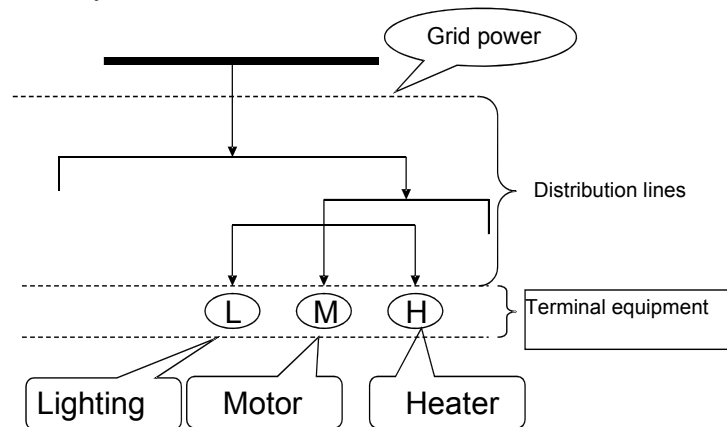


Fig.2-1 Power consumption of a company

Before devising a strategy to cut electricity use, an energy manager must answer **three questions**:

- » How and where are the power outlets that consume energy and how much is consumed at each outlet?
- » How much should each power outlet consume?
- » How can energy consumption be reduced?

To answer these questions, it is necessary to measure power consumption, and to understand **how and when power is consumed**.

Direct consumption is energy consumed during manufacturing. **Common use**

consumption is energy consumed during office hours. **Auxiliary energy consumption**, is additional energy consumed apart from central production processes and includes standby power generation, and other backup systems. (See illustration below).

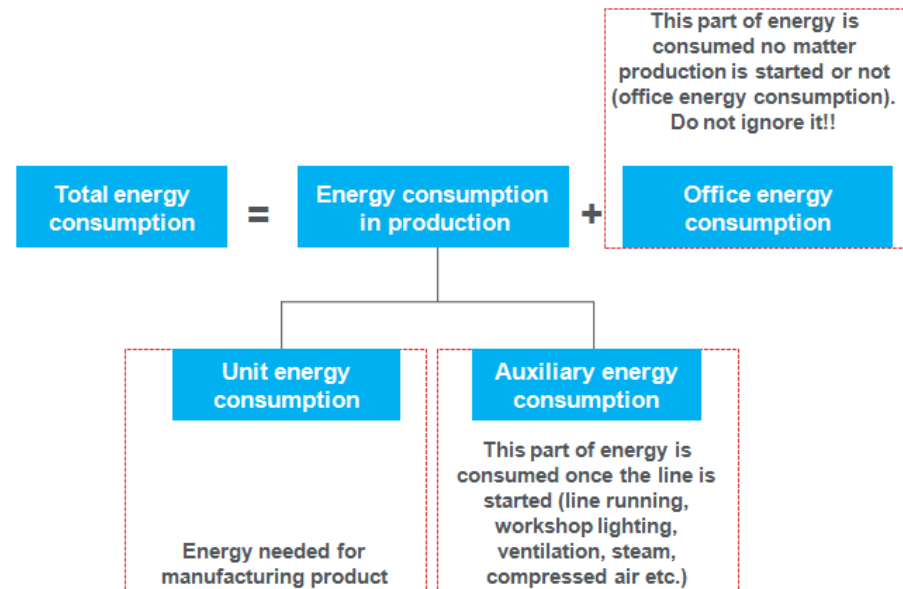
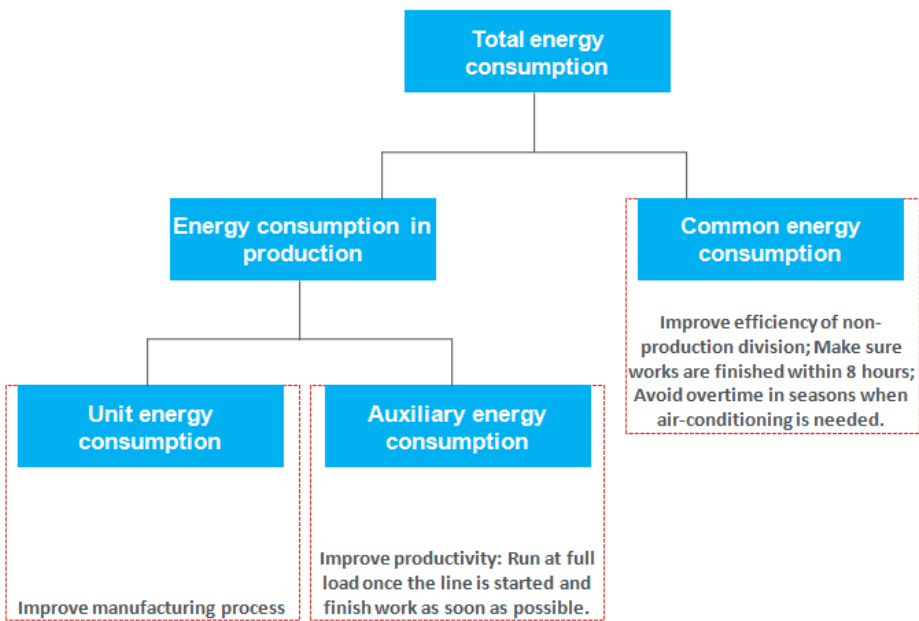


Fig. 2-2 Energy structure

Fig. 2-3 How to Save Energy



2.2 ENERGY MEASUREMENT

Measuring energy data is a key component for implementing an energy management plan. To be done effectively a manager needs to:

- » Establish an energy data measurement system;
- » Designate staff in charge of monitoring energy use;
- » Ensure proper use of energy measuring instruments and ensure safety of meters; and
- » Provide energy data in a complete and timely manner.

Designating staff members to implement energy measurement is important to creating an effective system.

Preparing a document that details ways the company measures energy is essential. This document includes information on monitoring data, as well as the staff members responsible for monitoring energy supply and consumption.

It also contains information on the instruments used to monitor energy consumption, details on storage, disposal, purchase, maintenance and calibration. It should also include a list of all equipment that measures energy consumption, detailing the name, model, manufacturer, serial number and date of calibration of the monitoring equipment (See below).

Larger companies usually have an independent energy measurement division or designated personnel in charge of energy measurement, while smaller companies designate an energy manager to conduct the task of measuring energy consumption.

List of Energy Measuring Instruments									
No.	Name	Model	Precision degree	Scope of measurement	Manufacturer	Delivery number	Management number	Location of installation (workshop, line, equipment) and usage (energy measurement, self-inspection, energy quality analysis)	Condition (qualified/allowed to use/forbidden to use)
1	Power meter	DX863-2K	0.5	—	Swiss	67877983	EP-301A01	High-voltage room, 1F, 301 Substation: Shenhua No.1 line	Qualified
2	Power meter	DS862-2	0.5	—	Wuxi Weida	1685	EP-301A03	High-voltage room, 1F, 301 Substation: Shenhua	Qualified
3	Vortex flowmeter	DN200	2.5	—	Shanghai Henghe	9803106	FQ-1100	152 pipe rack, Meiya Steam	Qualified
4	Electromagnetic flow meter	MAG-XE	0.5	—	ABB		WA-001	103C South Gate, tap water inlet of the factory	Allowed to use
5	Turbine flow meter	DN200	1	—	Shanghai Henghe		FQ-1350A	152 pipe rack, Meiya Softwater	Qualified
6									

Although measuring energy data only reflects "overall consumption" rather than energy efficiency, it is still fundamental to evaluating energy efficiency and

energy-saving opportunities.

Companies should measure energy consumption by taking into account:

- » Overall energy consumption.
- » Energy consumption of power lines and departments
- » Energy consumption of office and production equipment.

Alternatively, companies can measure power by:

(1) Measuring average power over a period of time:

- » For a system with constant power: Power consumption = actual power X working hours
- » For a system with variable power: Power consumption = average power X working hours

(2) Deciding where to install an electric power meter depends on whether the:

- » Equipment consumes large amounts of power or power consumption fluctuates;
- » Systems require modification to make them more energy efficient; and
- » Lines/departments that are subject to energy assessment and evaluation.

2.3 ENERGY-CONSUMING EQUIPMENT

Energy managers should identify office and production equipment that consume the most energy. According to the Principle for Equipping and Managing the Measuring Instrument of Energy Units (GB17167-2006), these pieces of equipment often consume power that is above the limit specified in Table 2-1.

Table 2-1 Limits of power consumption (or power) of main energy-consuming equipment

Energy type	Electricity	Coal	Crude oil, product oil, LPG	Heavy oil Residual oil	Gas Natural gas	Steam Hot water	Water	Others
Unit	kW	t/h	t/h	t/h	m ³ /h	Mw	t/h	GJ/h
Limit	100	1	0.5	1	100	7	1	29.26

A company can compare Table 2-1 and the criteria below against their own office and production equipment. Equipment is problematic if it:

- » Has estimated annual energy consumption above 80,000 kWh;
- » Is "major" equipment (large quantity, low unit power); and/or
- » Is not operating optimally in accordance with its design.

It is important for managers to hire experts to conduct regular energy efficiency tests on equipment that uses high amounts of energy (See Table 2-2). The tests should be carried out on as many pieces of equipment as possible so managers can gauge the overall energy needs of the company.

Table 2-2 Sample Energy Efficiency Test Form

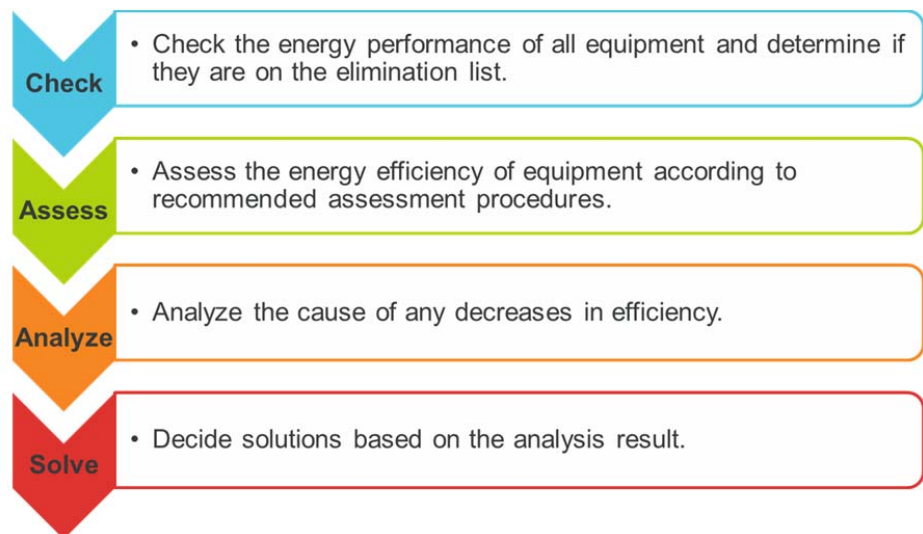
No.	Characteristics	Equipment Name	Quantity	Location	Tested efficiency	Design efficiency
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1	Over 500 lamps with annual lighting time more than 4,000 hours					
2	Over 1,000 lamps with annual lighting time more than 2,000~4,000 hours					
3	Annual power consumption of equipment >80000kWh					
4	Annual power consumption of equipment 20000—50000kWh Large deviation from design condition					
5	Boiler					

3. Energy Diagnosis Methods

3.1 MEASURING THE ENERGY EFFICIENCY OF COMPANY EQUIPMENT

Managers can measure the efficiency of all equipment using the following steps:



Potential challenges to conducting energy efficiency tests

Tests are expensive because they have to be conducted by experts and senior executives might be reluctant to replace equipment if there is insufficient data.

Managers may be biased if low equipment efficiency is due to bad management yet it is difficult to evaluate the investment return of such tests.

Check: Managers should identify whether their equipment is listed under the Guiding Catalogue of Industrial Structure Regulation (2011) or the Outdated Mechanical and Electrical Equipment (Products) with High Energy Consumption (first batch). Many companies are unwilling to retire old and outdated equipment, especially if it still works, not realizing that such equipment causes companies to incur additional costs in the long run.

Assess: Managers should also evaluate equipment that is not yet outdated to determine its energy efficiency based on national and international standards.

Analyze: Low operating efficiency is caused by:

- 1 Low quality equipment, including obsolete equipment that is listed in the “Eliminated Products Catalog”;
- 2 Improper service maintenance and operation; and

- 3 Process requirements that render equipment less efficient.

Solve: To correct efficiency problems managers should:

- 4 Replace obsolete equipment with new energy efficient models;
- 5 Improve energy management, including the establishment of equipment operation and maintenance policies; and
- 6 Integrate energy-saving technologies into the manufacturing process at a system level.

Case study of an electronics factory

At an assembly workshop in an electronics factory, 80 percent of total energy consumption goes towards auxiliary uses. This is largely because the workshop runs 8-hour shifts even though productivity is low and there are insufficient orders.

Suggestions:

Maintain higher productivity levels with the aim of completing production in six hours rather than eight. This will save the company 276 kWh of power (RMB 220) every day.

The remaining hours can be used to train operators.

Switch to night shifts as this could help reduce air-conditioning costs, since the price of electricity is significant lower at the night than in the daytime. (allowing for a reduction of 80 percent of energy costs.)

3.2 ENERGY STRUCTURE EVALUATION

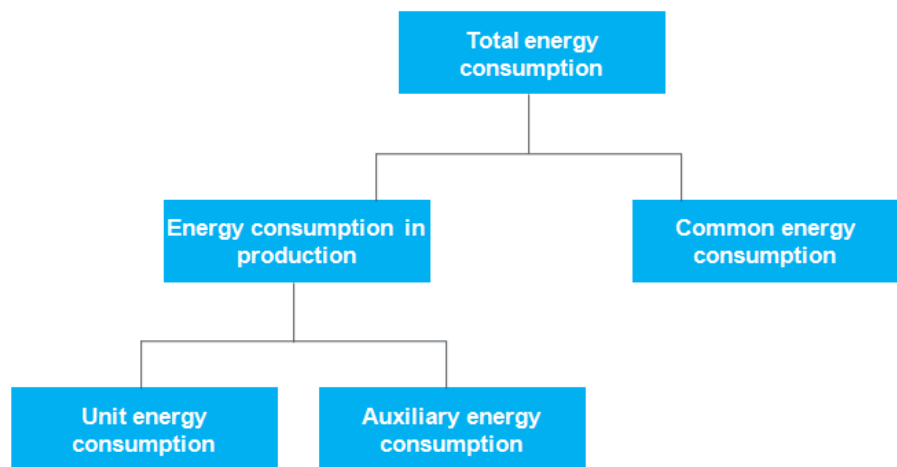


Fig. 2-4 Energy structure

As shown in the figure above, direct consumption, auxiliary consumption, and common consumption are independent parts of overall energy consumption. Reducing any of these three types of consumption would reduce overall energy consumption.

Auxiliary energy consumption energy-saving solutions:

- » Maintain high-efficiency production
- » Improve energy efficiency of high cost auxiliary equipment
- » Reduce working hours

Direct energy consumption energy-saving solutions:

- » Reducing direct energy consumption is relatively difficult. Generally, it can be done by reducing temperature, pressure, speed and flow of the production process.
- » Conduct a systematic production process analysis.

Common energy-saving solutions:

- » Increase work efficiency
- » Reduce overtime if possible
- » Create awareness of energy-saving management

3.3 SYSTEM ANALYSIS METHOD

Managers can conduct a system analysis of their energy systems to determine how these systems interact. For example, a manager might look at how electrical and water supply systems, function with the company's energy system as a whole. (Please see the case study below.) The aim is to identify energy-saving opportunities by identifying how different sets of electrical systems function together. The disadvantage of using this method is that testing requires advanced technical equipment and high-level technical expertise, which most SMEs would be unable to afford.

Case study: Waste water Treatment Plant Pump

A pump in a wastewater treatment plant (motor power: 11kW, flow: 100m³/h, head: 25m) had a measured efficiency rating of 95 percent and measured input power of 8.25kW. Although appearing stable, an inspection revealed that a valve used for satisfying the need of discharge flow (100 m³/h), was only opened by 1/3. After the valve was fully opened, the flow exceeded 10 percent of what was needed, and the efficiency dropped to 64 percent.

From an efficiency point of view, any non-fully opened valve in the pipeline is considered a waste of energy. The solution was to replace the pump with another that had a 100 m³/h flow, 15m head, 7.5kW motor power, and 6kW input power. Annual energy savings = $2.25 \times 24 \times 350 = 18,900$ kWh. The investment was paid back in half a year.

3.4 CLASSIFICATION METHODS

Managers seeking to reduce energy consumption can identify different ways that their companies can save energy. They can do this by using a classification method, which highlights (into a line diagram, as below) areas where the company can improve their use of energy, making energy waste easier to address.

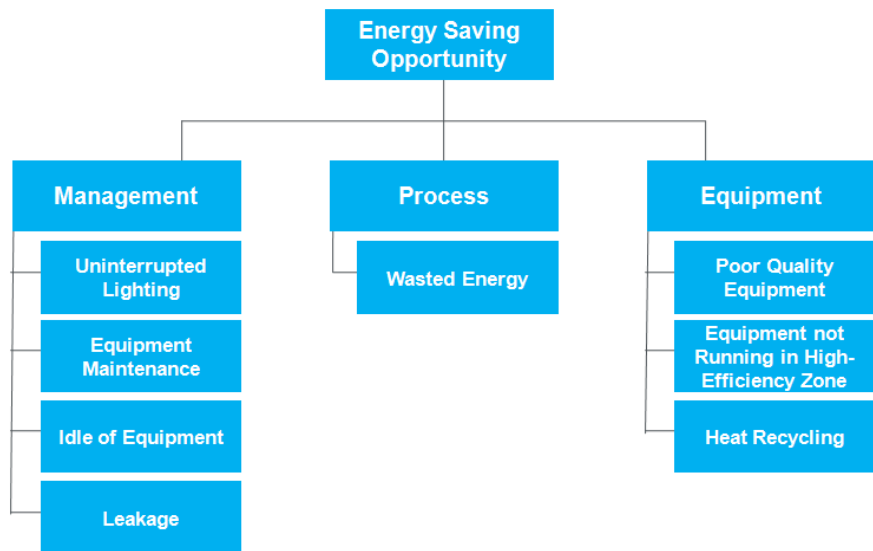


Fig. 2-5 Classification Method

3.4.1 REDUCING WASTE

To curb energy waste, managers must identify potential areas of waste, including:

- » Unnecessary uninterrupted lightening;
- » All kinds of leakage;
- » Equipment that consumes energy during non-working hours;
- » Times when production capacity falls below equipment capacity; and
- » Equipment inefficiency as a result of poor maintenance.

How can companies reduce energy waste through better management?

- » Managers should establish a corporate culture that encourages the elimination of wasted energy;
- » A patrol team can be set up to identify energy waste throughout the company;

- » Conduct energy-saving training; and
- » Implement incentive policies to award employees that conserve energy.

3.4.2 PROCESS ENERGY SAVING

Managers may also implement a process energy saving method to increase energy efficiency. This aims to optimize production processes by measuring areas where the company wastes energy, such as;

- » Over-lighting (over-lighting in a workshop of a Fortune-500 company is typically around 400,000 kWh);
- » Excessive use of heat or pressure. One well-known company, for instance, its energy use by 40 percent by reducing the amount of compressed air pressure it used for cleaning parts to 0.3MPa from 0.58MPa;
- » Unnecessarily low temperatures. A food processing plant, for example, saved 10 percent on its air-conditioning costs (RMB 200,000/year) by increasing its workshop temperature from 22 degree Celsius to 24 degree); and
- » Excessive consumption of high quality water or compressed air.

Why are there several opportunities to enhance energy savings in the production process?

In most cases, production process parameters for equipment are designed to satisfy the "maximum production load." But when it comes to actual production, the loads are usually processed slower, making parameter designs redundant. Equipment manufacturers often fail to consider energy saving in their designs, causing equipment to consume more energy than is necessary.

Philosophy for process energy savings: Any energy supply with unnecessary output, no matter how efficient, is a waste.

Why process energy saving has the highest effect: To reach the terminal, energy first passes through the transformer, conversion devices and the distribution equipment.

Input energy = output energy/ η

η : system efficiency. Since η is less than 1, by reducing output energy, it is able to achieve a significant energy saving effect. For example, if the output of a compressed air system with an efficiency of 0.3, comprising five components, including power supplies, compressors, air processing equipment, distribution pipelines and pneumatic tools, is reduced by 1kW, input energy will fall by 3.33kW -- a significant energy saving.

Process energy saving is a low cost way of improving the quality of products, while reducing energy consumption. To implement this process, managers should:

- » Identify how much energy each business process consumes and find ways to control the system's parameters, including the air temperature, atmospheric pressure, etc.
- » Use a team of engineers to control energy flows through production systems. As energy flows from higher temperatures to lower temperatures, it creates heat flows; and it moves from higher pressure to lower pressure, it causes expansion.

3.4.3 MAXIMIZING THE EFFICIENCY OF EQUIPMENT

There are two key issues to consider when trying to save energy:

- » Design efficiency reflects maximum efficiency and cannot be achieved in actual production.
- » Energy-saving technology resolves the issue when equipment is "operating in the low efficiency zone", but it is not a remedy for poorly maintained equipment. Equipment maintenance is thus vital and should take place on a regular basis, as per the requirements of its maintenance manual.

To enhance energy saving on equipment, managers should:

- » Inspect the condition of equipment regularly to avoid energy waste caused by poor maintenance and avoid: (a) the loss of heat transfer in heat exchangers; (b) blockages in the fuel filters; and (c) overloading of lubrication systems, among other issues. One company found a filter in a compressed air system was blocked, causing a 146kW overload of the power generator. The air generated by the compressor, meanwhile, only amounted to 18m³/min.
- » Consider whether the design specifications of the company's equipment causes low energy efficiency, and if that is the case, whether the company can replace it with one that is a better fit for its needs.
- » As an alternative to replacing equipment, companies can install inverters or other energy-saving devices. Although doing this saves energy to some extent, it does not reduce costs or overall energy consumption. The example below demonstrates this point.
- » Consider adding a frequency converter (see side bar) to increase the frequency cycle of fans or pumps operating in low efficiency zone; to improve systems with fluctuating pressures or with compressors running constantly.

Frequency Conversion Devices

Equipment that has a driver motor can reach its intrinsic efficiency only when it runs under conditions for which it was ideally designed.

When selecting equipment, companies usually consider the maximum power required, but most of the time, the equipment doesn't run at its maximum power and so is not as efficient as it could be.

This is where frequency conversion technology comes in to play. Frequency conversion technology can make fluid machines run close to their design efficiency, but this technology cannot make the efficiency of motor systems any higher.



Examples of Frequency Conversion



Before



After

Fig. 2-7 Frequency conversion and constant voltage control project for the power station of a company

A company invested RMB 50,000 in a frequency conversion and constant voltage control project for its power station. The new system reduced power consumption by 6500 kWh and saved the company RMB 5,200 a month.



Fig. 2-8 Elevator network controller

By introducing an elevator network control system based on frequency conversion technology, a company saved 6,000 kWh power per month, equivalent to monthly savings of RMB 4,800 (based on a price of RMB 0.8/kWh).